

Aprendizaje de Máquina

U3 Otros Métodos de Agrupamiento

Héctor Maravillo

Section 1

OPTICS

OPTICS

An **extension** of DBSCAN that overcomes the necessity of choosing carefully the parameters ϵ and q is the **OPTICS (Ordering Points To Identify the Clustering Structure)** algorithm.

This generates a **density-based cluster ordering**, representing the **intrinsic hierarchical cluster structure** of the data set in a comprehensible form.

Experiments indicate that the run time of OPTICS is roughly equal to 1.6 of the runtime required by DBSCAN.

On the other hand, in practice one has to run DBSCAN more than one time for different values of ϵ and q .

Motivation

An important property of many real-data sets is that their **intrinsic cluster structure** cannot be characterized by **global density parameters**.

Very different local densities may be needed to reveal clusters in different regions of the data space. For instance, see the following figure:

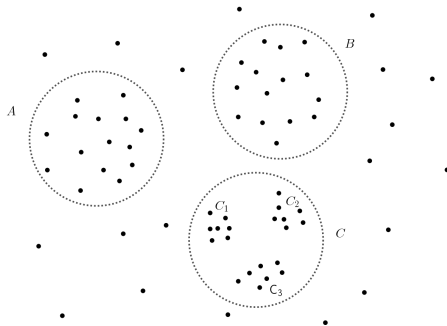


Figure: Clusters with different density parameters

Density-based clustering

The key idea of **density-based clustering** is that for each object of a cluster the neighborhood of a given radius (ϵ) has to contain at last a minimum number of objects (*MinPts*), i.e., the cardinality of the neighborhood has to exceed a threshold.

However, the most of this kind of algorithms (such as DBSCAN) has a **weaknesses**: the problem of detecting meaningful clusters in data of **varying density**.

Density-based Cluster-Ordering

For a constant *MinPts*-value, density-based cluster with respect to a higher density (i.e. a lower value for ϵ) are **completely contained** (**nested**) in density-connected sets with respect to a lower density (i.e. a higher value for ϵ).

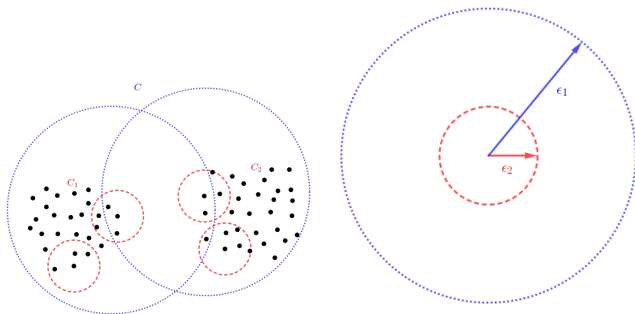


Figure: Illustration of **nested** density-based clusters: C_1 and C_2 are density-based clusters with respect to $\epsilon_2 < \epsilon_1$ and C is a density-based cluster with respect to ϵ_1 completely containing the sets C_1 and C_2 ($MinPts = 3$).

Density-based Cluster-Ordering

Consequently, we could **extend the DBSCAN algorithm** such that several distance parameters are processed at the same time, i.e, the density-based clusters with respect to **different densities** are **constructed simultaneously**.

To produce a consistent result, however, we would have to obey a **specific order** in which objects are processed when expanding a cluster.

We always have to select an object which is density-reachable with respect to the lowest ϵ value to guarantee that clusters with respect to higher density (i.e. smaller ϵ values) are finished first.

OPTICS algorithm

The **OPTICS algorithm** (**O**rding **P**oints **T**o **I**dentify the **C**lustering **S**tructure) works in principle like such an **extended DBSCAN algorithm** for an infinite number of distance parameters ϵ_i which are smaller than a “**generating distance**” ϵ (i.e. $0 \leq \epsilon_i \leq \epsilon$).

The only difference is that we do not assign cluster membership. Instead, we store the **order** in which the objects are processed and the information which would be used by an extended DBSCAN algorithm to assign cluster membership.

This information consist of only two values for each object:

- The **core-distance**
- A **reachability-distance**.

Definitions

Let $\mathbf{x}$ be an object from a dataset $\mathbf{X}$, let ϵ be a distance value, let $N_\epsilon(\mathbf{x})$ be the ϵ -neighborhood of $\mathbf{x}$, let $MinPts$ be a natural number and let $MinPts$ -distance($\mathbf{x}$) be the distance from $\mathbf{x}$ to its $MinPts'$ neighbor.

Definitions

The **core-distance** of $\mathbf{x}$ is defined as

$$cd_{\epsilon, MinPts}(\mathbf{x}) \begin{cases} \text{undefined} & \text{if } |N_\epsilon(\mathbf{x})| < MinPts \\ MinPts\text{-distance}(\mathbf{x}) & \text{otherwise} \end{cases}$$

In other words, it is simply the **minimum distance** $\epsilon' < \epsilon$ for a data point $\mathbf{x}$ to be considered as a **core point**.

Definitions

Definitions

The **reachability-distance x with respect to object y** is defined as

$$rd_{\epsilon, MinPts}(\mathbf{x}, \mathbf{y}) = \begin{cases} \text{undefined} & \text{if } |N_{\epsilon}(\mathbf{y})| < MinPts \\ \max \{ \text{core-distance}_{\epsilon, MinPts}(\mathbf{y}), d(\mathbf{x}, \mathbf{y}) \} & \text{otherwise} \end{cases}$$

Intuitively, the reachability-distance of an object $\mathbf{x}$ with respect to another object $\mathbf{y}$ is the smallest distance such that $\mathbf{x}$ is directly density-reachable from $\mathbf{y}$ if $\mathbf{y}$ is a core object.

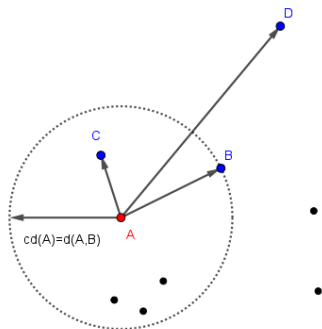
This concept is introduced to ensure that points in less dense regions (further from the core) can still be reached and included in a cluster. It's a measure that respects the density of the region around point $\mathbf{x}$ while recognizing the actual distance to point $\mathbf{y}$.

Definitions

Consider the following distance $d(A, B) = 2$, $d(A, C) = 1.18$ and $d(A, D) = 4.47$, with $MinPts = 5$

So the core-distance of A is 2, while the reachability-distance are:

- $rd(B, A) = 2$
- $rd(C, A) = 2$
- $rd(D, A) = 4.47$



Remarks

Both core-distance and reachability-distance are undefined if no **sufficiently dense cluster** (w.r.t. ϵ) is available.

Given a sufficiently large ϵ , this never happens, but then every ϵ -neighborhood query returns the entire database, resulting in $O(n^2)$ runtime.

Hence, the ϵ parameter is required to **cut off the density of clusters** that are no longer interesting, and to speed up the algorithm.

The parameter ϵ is, strictly speaking, not necessary. It can simply be set to the maximum possible value.

Remarks

Without any index support, to answer such a region query, a scan through the whole database has to be performed. In this case, the run-time of OPTICS would be $O(n^2)$.

If a tree-based spatial index can be used, the run-time is reduced to $O(n \log n)$ since region queries are supported efficiently by spatial access methods such as the R^* -tree or the X -tree.

Pseudocode

The OPTICS algorithm generates the **augmented cluster-ordering** consisting of the ordering of the points, the reachability-values and the core-values.

Specifically, given a dataset $\mathbf{X}$, the OPTICS algorithm generates an ordering of the points $o : \{1, \dots, n\} \rightarrow \mathbf{X}$ and corresponding reachability-values $r : \{1, \dots, n\} \rightarrow \mathbb{R}_{\geq 0}$.

Algorithm

OPTICS begins with an empty seed-list and picks a random point p , finds its ϵ -neighborhood, and **determines its core distance**.

The **reachability distance** of this first point is **set to undefined**, and current point p is written to the output list.

- If point p is not a core point, then the algorithm simply picks another random point from the input dataset.
- If point p is a core point, then the **reachability distance** of **each neighboring point q** with respect to point p **is calculated**.

All neighboring points are then inserted into the seed-list, and the list **is sorted** in ascending order by the reachability distance value.

Algorithm

In the next iteration, OPTICS takes the point that is at the top of the seed-list and, in case it is a core point, finds its ϵ -neighborhood, determines its core distance, and calculates the reachability distance for each of its neighboring points.

- If the newly calculated reachability distance of some unprocessed point is smaller than the one present in the seed-list, the value is updated to a smaller value, and the seed-list is sorted.
- If the current point is not a core point, then OPTICS moves to the next point from the seed-list.

The process continues until each point is processed and the seed-list is empty.

Reachability plot

The **ordering output** from the OPTICS algorithm can be visualized using a **reachability plot** which is a special form of a **dendrogram**.

It is a 2D plot with points in the **ordering** returned by the OPTICS algorithm on the **x-axis** and their **reachability distances** on the **y-axis**.

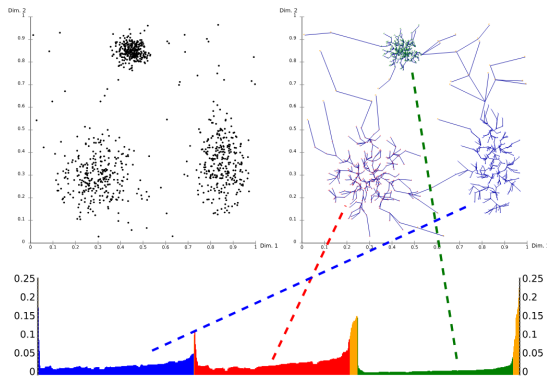


Figure: Reachability plot

References

- Theodoridis, S. & Koutroumbas, K. *Pattern Recognition*, 4th ed., Academic Press, 2009